

# Can Large Language Models Predict 3PL Item Parameters in Brazilian Portuguese? An Exploratory Study with ENEM Items

Pedro Otavio Mantovani<sup>1</sup>

Advisor: André Carlos Ponce de Leon Ferreira de Carvalho<sup>2</sup>

University of São Paulo, Institute of Mathematics and Computer Science (ICMC) SP, Brazil

pedro.mantovani6@usp.br<sup>1</sup>, andre@icmc.usp.br<sup>2</sup>

## Abstract

**3 commercial Large Language Models** were used to estimate the **3 logistic parameters** of items from the Brazilian National High School Exam (ENEM). The models could partially explain **item difficulty** (reaching  $r = 0.69$ ), but failed to recover the other two.

**Keywords:** Large Language Models (LLMs); Item Response Theory (IRT); item parameter estimation.

## Background

- Item calibration under **Item Response Theory (IRT)** traditionally requires **pilot samples**, which leads to **high costs and risk of item leakage**. The ENEM itself is scored using the **3PL model**.

- Prior work on automated item-parameter prediction has mostly targeted **item difficulty**, using traditional NLP/ML pipelines (Belov et al., 2022; Benedetto, 2023), with results ranging from  $r = 0.4$  to  $0.79$  in certain cases.

- Almost all works on the area tests **English-language assessments** (Ferrara et al., 2022).

- Large Language Models (LLMs) offer a new, low-cost avenue for estimating the parameters and are not very explored yet. One of the uses explored was simulating examinee response patterns under a 2PL model (Maeda, 2025).

## Objective

To investigate whether state-of-the-art commercial LLMs can estimate complete **3PL item parameters (a, b, c)** directly from the textual content of Brazilian Portuguese assessment items, using a **few-shot prompting** approach.

## Method

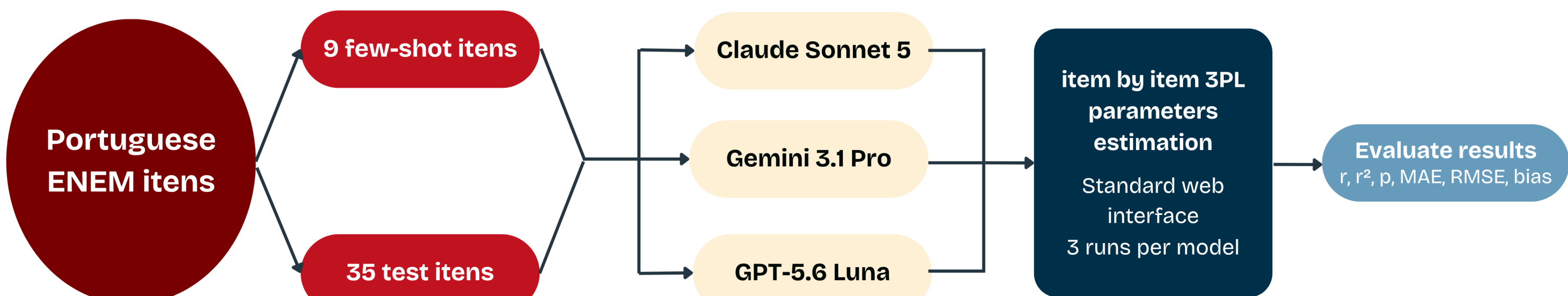


Figure 1: Method scheme

## Results

- Difficulty (b)** was the only parameter with a **consistent, above-chance signal**

- Pearson correlation with official values ranged from **0.37 to 0.69** across models for b estimation, which is still insufficient for operational calibration, but is non-random

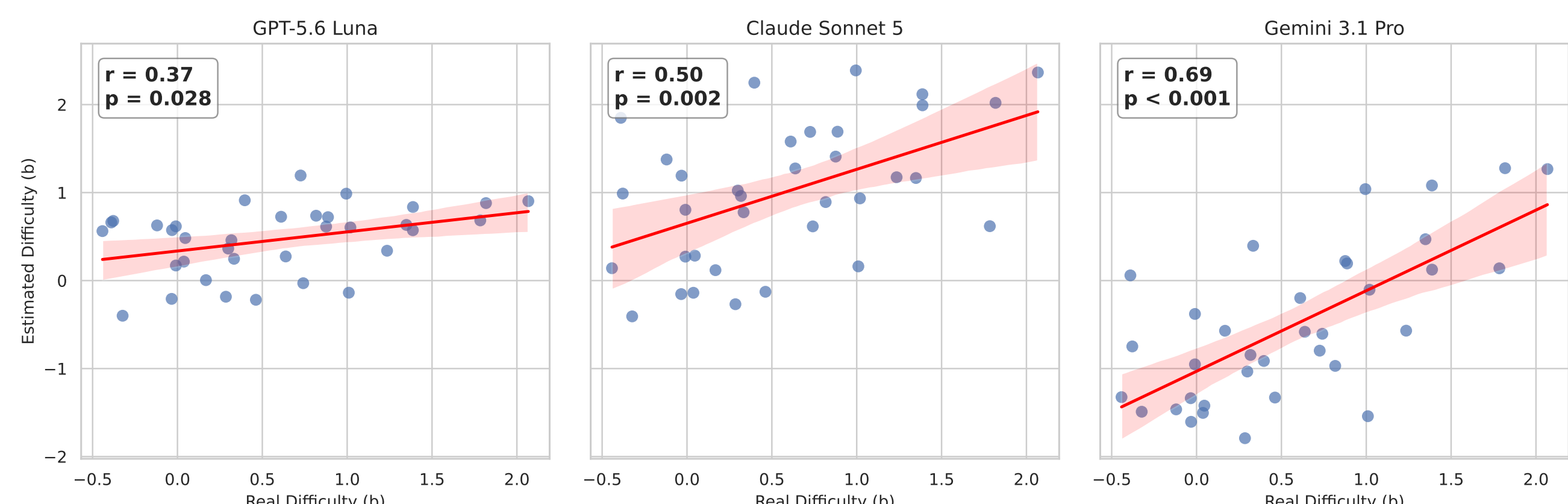


Figure 2: Average model difficulty (b) estimation vs. real parameter.

- Discrimination (a)** and **guessing (c)** showed **near-zero/null predictive power**, with a marked **regression-to-the-mean pattern**

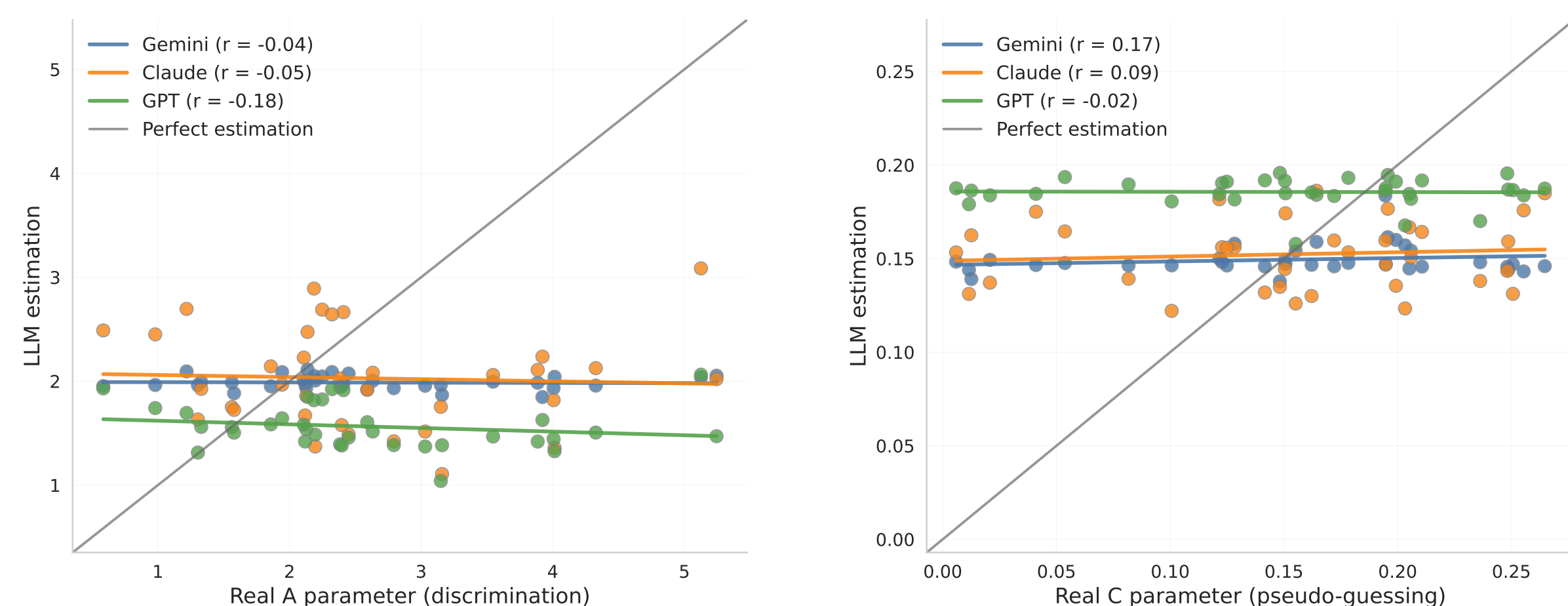


Figure 3: Average model discrimination (a) and pseudo-guessing (c) estimation vs. real parameter.

- Although a and b show essentially no correlation in the official calibration, the LLM-estimated a and b values were **highly intercorrelated (0.33 – 0.94)**, suggesting a shared, possibly difficulty-driven heuristic rather than independent estimation of the two parameters

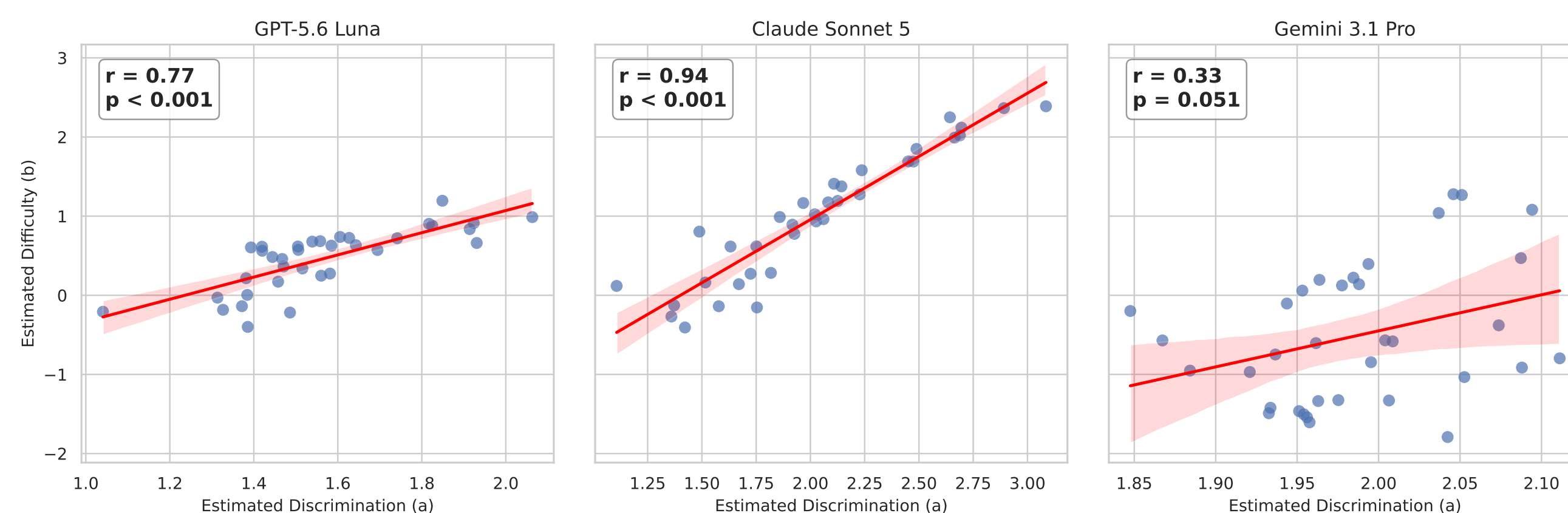


Figure 4: Correlation between estimated a and b parameters.

- Stability and ensembling:** Between-run agreement, measured via Intraclass Correlation Coefficient (ICC), ranged from **0.60 to 0.93** across models. Despite this variability, averaging estimates across the three runs mitigated noise and prevented poor individual generations from degrading results. The resulting **ensemble matched or outperformed the best single run**, without requiring ground-truth data to select which run to trust.

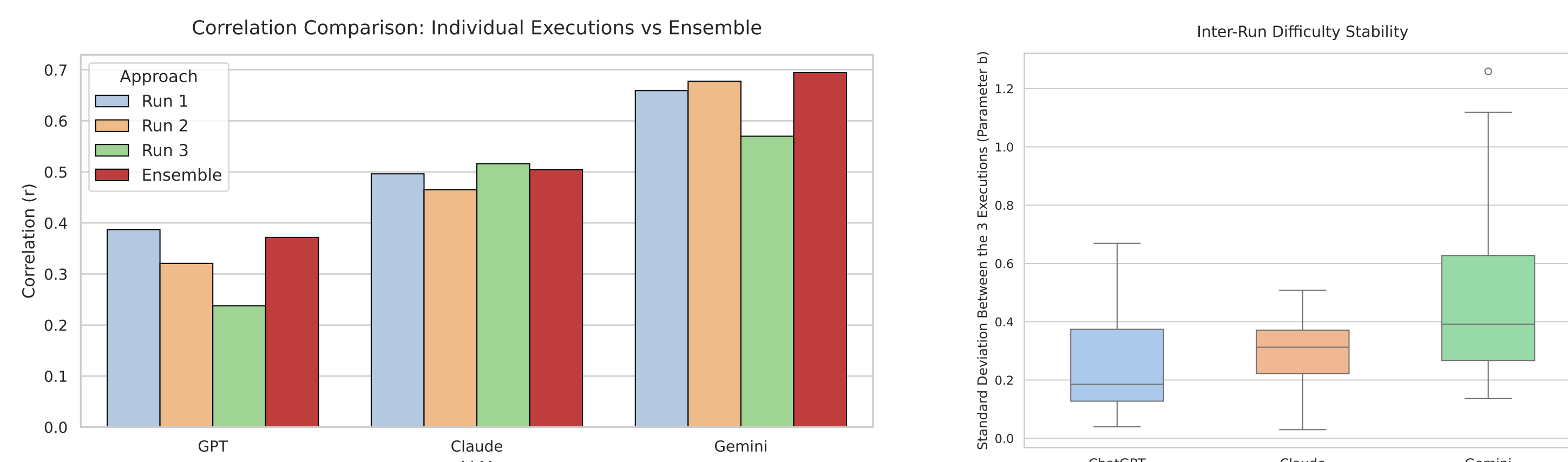


Figure 5: Comparison between r values for single runs and average value.

Figure 6: Standard Deviation between the three runs.

## Discussion

Despite being an exploratory, small-scale study, using off-the-shelf, non-fine-tuned, general-purpose commercial LLMs, in Portuguese (a comparatively lower-resource language for these systems), with only 9 few-shot examples, the difficulty-prediction performance already **falls within the range reported by prior studies** using dedicated NLP/ML pipelines trained specifically for item-difficulty prediction. The comparatively weaker recovery of the other parameters was also seen in other studies.

This suggests LLMs are already capturing psychometrically relevant signal “for free,” without any task-specific training, a **meaningful baseline for future AI-assisted psychometrics**.

The  $a, b$  correlation at estimations gives a plausible explanation for why  $a$  shows no predictive power despite  $b$  showing a moderate one. This suggests that another approach for estimating  $a$ , might have better results.

## Limitations & Future Work

- Small test set** (35 items) and few-shot example set (9 items), restricted to a single content domain (language items) → evaluate on larger, calibrated item sets spanning multiple domains (e.g., mathematics, sciences) to disentangle sample-size from domain effects

- Near-zero performance for a and c**, with a marked regression-to-the-mean pattern → test whether larger few-shot sets or fine-tuning reduce this pattern, before concluding these parameters are unrecoverable from text alone

- Unstructured, non-explanatory prompting** → adopt structured, explainable prompting (e.g., asking the model to justify each estimate) to diagnose systematic errors such as the  $a-b$  collinearity observed here

- Single language and exam** (Portuguese, ENEM) → compare against English and other language items to compare its impact on estimation

- General-purpose, off-the-shelf models only** → assess how much more paid or specialized models can improve of the estimations.

## Conclusion

Contemporary LLMs can already capture some psychometrically relevant information from Brazilian Portuguese assessment content, particularly for **item difficulty**, at levels broadly comparable to specialized NLP/ML approaches, despite an exploratory, small-scale, non-fine-tuned setup. Results are not yet sufficient for operational item calibration, but they support continued investigation of LLMs as a **low-cost complement to traditional IRT calibration**.

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Full references, prompts, code & results:

[github.com/pedro-mantovani/IACAT\\_repo](https://github.com/pedro-mantovani/IACAT_repo)